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CSCI E-102
Assignment 1, Part 1

We want to investigate whether the smell of fresh baking at a bakery has a causal effect on sales. Over 10 days when all the bakery's windows are open, we observe the sample mean and standard deviation of sales are $\bar{X}_1 = \$1320$ and $S_1 = \$22$, respectively. Similarly, over 12 days when all the windows are closed, we get $\bar{X}_2 = \$1220$ and $S_2 = \$10$. Assuming samples come from normal populations, use t -statistic to test if there is a difference in expected sales, μ_1 and μ_2 . Answer in steps:

1. Give appropriate H_0 , H_1 . Assume one-sided test.
2. Calculate the test statistic.
3. Give the p-value. What is the conclusion at $\alpha = 5\%$ level of confidence?

Hint: The sample sizes are small, so we cannot use Normal approximation. The test statistic follows t -distribution with the number of degrees of freedom given by:

$$\text{d.f.} \approx \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{S_1^2}{n_1}\right)^2 + \frac{1}{n_2-1}\left(\frac{S_2^2}{n_2}\right)^2} = \frac{\left(\frac{22^2}{10} + \frac{10^2}{12}\right)^2}{\frac{1}{10-1}\left(\frac{22^2}{10}\right)^2 + \frac{1}{12-1}\left(\frac{10^2}{12}\right)^2} \approx 12$$

4. Please discuss whether there is causal effect of the smell of fresh baking on sales. Consider what else could be different on those days when the windows are open and closed. Can one improve the experiment? Please be specific.

SOLUTION:

1) $H_0 \leq \$1,320$
 $H_1 > \$1,320$

2) The test statistic is:

$$TS = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{1320 - 1220}{\sqrt{\frac{22^2}{10} + \frac{10^2}{12}}} = \frac{100}{\sqrt{\frac{484}{10} + \frac{100}{12}}} =$$

$$\frac{100}{7.53} = 13.28$$

3) The p-value is:

$$p\text{-value} = P(T_{12} > 13.28) = 0.000000007756479$$

figured out by using R:

`pt(13.28, df=12, lower.tail=FALSE)`

4) There does seem to be a causal effect of the smell of fresh baking on sales, due to the fact that the p-value is less than 0.05 and extremely small (approaching 0).

You could improve the experiment by increasing the sample size and controlling for weather, which could be the driving factor behind whether the windows are open or not. Maybe people are in a better mood and therefore more inclined to buy on nice weather days?